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1.write Programs for summation of series $1+X+X^2+X^3+\dots$ with different time complexities.

Method 1:

```
#include<stdio.h>
#include<math.h>
int main(){
    int x,n,sum;
    printf("Enter the values of x and n: ");
    scanf("%d %d",&x,&n);
    if(x==1){
        sum=n+1;
    }else{
        sum=(pow(x,n+1)-1)/(x-1);
    }
    printf("The sum of the series is: %d",sum);
    return 0;
}
//time complexity O(1).
```

Method 2:

```
#include<stdio.h>
#include<math.h>
int main(){
    int x,n,i;
```

```

double sum=0;

printf("Enter the values of x and n: ");

scanf("%d %d",&x,&n);

for(i=0;i<=n;i++){
    sum+=pow(x,i);
}

printf("The sum of the series is: %lf",sum);

return 0;
}

//time complexity O(n).

```

Terminal process:

```

Enter the values of x and n: 5
5
The sum of the series is: 3906.000000

...Program finished with exit code 0
Press ENTER to exit console.

```

2. Write a program to find the Kth smallest element in the array.

```

#include<stdio.h>
int main(){
    int a[100],i,j,k,temp,n;
    printf("Enter the size of the array: ");
    scanf("%d",&n);
    printf("Enter the elements of the array: ");
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("Enter the value of k: ");
    scanf("%d",&k);
    for(i=0;i<n-1;i++){
        for(j=i+1;j<n;j++){
            if(a[i]>a[j]){

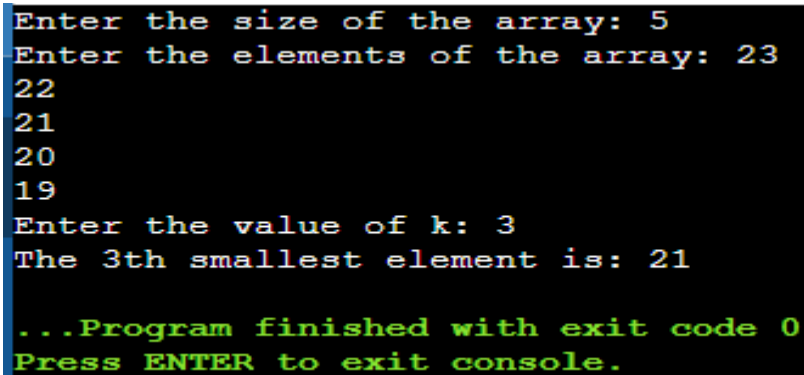
```

```

        temp=a[i];
        a[i]=a[j];
        a[j]=temp;
    }
}
printf("The %dth smallest element is: %d",k,a[k-1]);
return 0;
}
//time complexity is O(n^2)

```

Terminal Process:



```

Enter the size of the array: 5
Enter the elements of the array: 23
22
21
20
19
Enter the value of k: 3
The 3th smallest element is: 21

...Program finished with exit code 0
Press ENTER to exit console.

```

3. Create a Binary Search Tree and perform the insertion, deletion operations.

Program:

```

#include <stdio.h>

#include <stdlib.h>

struct node {
    int data;
    struct node *left;
    struct node *right;
};

struct node* createNode(int data) {

```

```

    struct node* newNode = (struct node*)malloc(sizeof(struct node));

    newNode->data = data;

    newNode->left = NULL;

    newNode->right = NULL;

    return newNode;
}

struct node* insert(struct node* root, int data)
{ if (root == NULL)
return createNode(data);
if (data < root->data)
    root->left = insert(root->left, data);
else if (data > root->data)
    root->right = insert(root->right, data);
return root;
}

struct node* findMin(struct node* root)
{ while (root->left != NULL)
    root = root->left;
return root;
}

struct node* deleteNode(struct node* root, int data) {
    struct node* temp;

```

```

if (root == NULL)
    return root;
if (data < root->data)
    root->left = deleteNode(root->left, data);
else if (data > root->data)
    root->right = deleteNode(root->right, data);
else {
    if (root->left == NULL) {
        temp = root->right;
        free(root);
        return temp;
    }
    else if (root->right == NULL) {
        temp = root->left;
        free(root);
        return temp;
    }
    temp = findMin(root->right);
    root->data = temp->data;
    root->right = deleteNode(root->right, temp->data);
}
return root;
}

void inorder(struct node* root) {

```

```

if (root != NULL) {
    inorder(root->left);
    printf("%d ", root->data);
    inorder(root->right);
}
}

int main() {
    struct node* root = NULL;

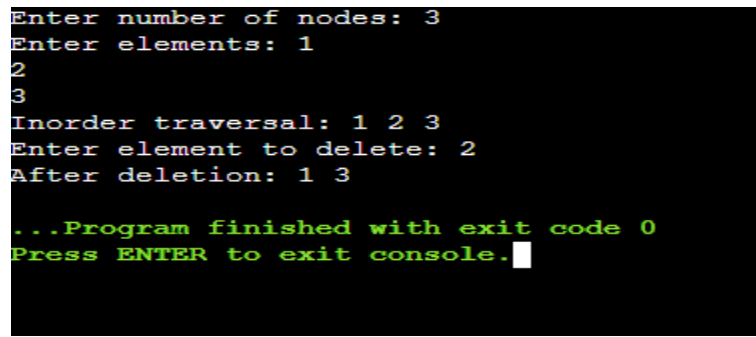
    int n, i, data, del;
    printf("Enter number of nodes: ");
    scanf("%d", &n);
    printf("Enter elements: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &data);
        root = insert(root, data);
    }
    printf("Inorder traversal: ");
    inorder(root);
    printf("\nEnter element to delete: ");
    scanf("%d", &del);
    root = deleteNode(root, del);
    printf("After deletion: ");
    inorder(root);
}

```

```
return 0;
```

```
}
```

Terminal Process:

A screenshot of a terminal window with a black background and white text. The text shows the execution of a program: 'Enter number of nodes: 3', 'Enter elements: 1', '2', '3', 'Inorder traversal: 1 2 3', 'Enter element to delete: 2', 'After deletion: 1 3'. Below this, in green text, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.' followed by a white cursor block.

```
Enter number of nodes: 3
Enter elements: 1
2
3
Inorder traversal: 1 2 3
Enter element to delete: 2
After deletion: 1 3

...Program finished with exit code 0
Press ENTER to exit console.█
```